

Algebra PhD Exam, May 1999

Please do 7 out of the 11 problems below.

1. An algebraic extension field F of K is said to be normal over K if every irreducible polynomial in $K[x]$ that has a root in F actually splits in $F[x]$. Prove that an algebraic extension F of K is normal over K if and only if for every irreducible $f \in K[x]$, f factors in $F[x]$ as a product of irreducible factors all of which have the same degree.
2. This problem has two parts.
 - (a). Show that the additive group of rationals $\mathbb{Q}$ is not free.
 - (b). Show that the group $\mathbb{Q}^+$ of all positive rationals (under multiplication) is free abelian with basis $\{p : p \text{ is prime in } \mathbb{Z}\}$.
3. Show that any simple group G of order 60 is isomorphic to A_5 . (If you want, you may use the fact that for each $n \geq 2$, A_n is the only subgroup of S_n of index 2.)
4. Let R be an integral domain and for each maximal ideal M , consider the localization R_M as a subring of the quotient field of R . Show that $\bigcap R_M = R$, where the intersection is taken over all maximal ideals M of R .
5. This problem has two parts.
 - (a). Define solvable group.
 - (b). Prove that any group of order 48 is solvable (without using Burnside's (p, q) -Theorem).
6. Let R be a ring. Show that the following conditions on an R -module P are equivalent.
 - (i). P is projective.
 - (ii). Every short exact sequence
$$0 \rightarrow A \xrightarrow{f} B \xrightarrow{g} P \rightarrow 0$$
is split exact.
 - (iii). There is a free module F and an R -module K such that $F \cong K \oplus P$.
7. Let C and D be categories, and let S and T be covariant functors from C to D .
 - (a). Define a natural isomorphism $\alpha : S \rightarrow T$.
 - (b). If B is a unitary left module over a ring R with identity, show that there is a "natural" isomorphism of modules
$$\alpha_B : R \otimes_R B \cong B.$$
8. Let R be a commutative ring with identity.
 - (a). Let $\mathcal{A}$ be an ideal in R , and assume that M is a finitely generated R -module such that $\mathcal{A}.M = M$. Show that there is some $a \in \mathcal{A}$ satisfying $(1 + a)M = 0$.
 - (b). Let M be a finitely-generated R -module. Show that $J(R).M = M$ implies $M = \{0\}$. (Here, $J(R)$ denotes the Jacobson radical of R .)

9. This problem has two parts.

- (a). Show that if R is a unique factorization domain, then R is integrally closed.
- (b). Find the integral closure of $\mathbb{C}[x^5, x^7]$.

10. This problem has two parts.

- (a). Show that $\mathbb{C}[x, y]/(xy - 1)$ (quotient of polynomial ring in 2 variables) is not isomorphic to $\mathbb{C}[t]$ (polynomial ring in 1 variable).
- (b). Show that the set

$$\{(m, n) : m, n \in \mathbb{Z}\}$$

is not an algebraic variety in $\mathbb{C}^2$.

11. This problem has three parts.

- (a). Find the Galois group of $K : \mathbb{Q}$, where K is a splitting field over $\mathbb{Q}$ for $t^4 + t^2 - 6$.
- (b). Write down the subgroups of the Galois group.
- (c). Write down the corresponding fixed fields.